

VAMSHI KRISHNA MADHAVAN

Dallas, Texas, USA | vamshi-madhavan@outlook.com | +1 (469) 315-4650 | vamshi0104.github.io/imVamshi
linkedin.com/in/vamshi0104 | github.com/vamshi0104

Skills

- **Programming & Frameworks:** Python, Java, C, SQL, PHP, JSP, Spring Boot, Spring Batch, Spring Data JPA, Flask, Streamlit, GraphQL, REST API, TensorFlow, Keras, PyTorch, scikit-learn, Pandas, NumPy, LangChain
- **Cloud & DevOps:** AWS (S3, EC2, ECS, Lambda, DynamoDB, RDS, CloudWatch, Kubernetes), Docker, CI/CD Pipelines, Jenkins, Git, Bitbucket, Apigee
- **Big Data & Databases:** Apache Spark, PySpark, Apache Airflow, MySQL, PostgreSQL, Redis, Memcache, Distributed Systems, ETL & Data Pipelines, Cloud-Native Microservices
- **Tools & Methodologies:** JIRA, Agile Practices, Scalable API Development

Experience

Experian – Texas, USA
Senior Software Engineer

December 2024 – Present
Promoted February 2025

- Architected and re-engineered a distributed big data processing platform, reducing end-to-end batch runtime from 85 minutes to 7 minutes (12× improvement) while delivering 12× throughput gains across analytics and ML pipelines via Spark DAG optimization, partition rebalancing, executor memory tuning, distributed PySpark refactoring, and Airflow orchestration optimization.

Software Engineer, Experian – Hyderabad, India

April 2021 – January 2024

- Architected migration of a high-volume batch microservice to a distributed Big Data platform, driving 700% TPS growth and 10× latency reduction across Spring Boot, Airflow, Redis, and AWS (S3, ECS, EMR).
- Designed and implemented the CensusHub analytical data warehouse, enabling scalable multi-million record processing through optimized partitioning, indexing strategies, and pipeline parallelism.

Software Engineer Intern, Cognitree – Bengaluru, India

December 2020 – April 2021

- Built enterprise-grade services while developing strong foundations in distributed systems, modular architecture, and API-driven design.

Projects

- **VocalVerse** | *Python, NLP*: Developed a PyPI package for text translation, speech synthesis, and audio transcription using OpenAI Whisper and Google Text-to-Speech.
- **Multilingual AI Chatbot** | *Python, NLP, ML*: Built a polyglot chatbot with LLaMA3 and OpenAI for text, voice interactions.
- **Time-Series Mining for Agriculture** | *Python, ML*: Designed an Active Learning framework for analyzing seasonal time-series data, enhancing crop monitoring.
- **Analysis of Clickstream Data for Online Shopping** | *Python, Machine Learning*: Conducted a comprehensive machine learning analysis of clickstream data, evaluating consumer behavior.
- **AR Math Solver** | *Python, ML, NLP, Flask, FastAPI*: Developed a math solver that extracts equations using image processing and LaTeX-OCR and generates step-by-step solutions via WizardMath LLM served through FastAPI.
- **URL Shortener Service** | *Spring Boot, Ehcache, MySQL*: Created a distributed URL shortener with efficient caching.
- **Vintage Knowledge** | *HTML, CSS, JavaScript, jQuery*: Crafted an educational website for learning 'C' programming. Offered a structured, user-friendly interface for comprehensive coding tutorials.
- **Roma Mart** | *HTML, CSS, JavaScript, jQuery, PHP, MYSQL*: Delivered an interactive retail store management system as a freelance project, deploying real-time inventory and sales management features to enhance operational workflows.

Education

University of Houston, Master of Science in Data Science

May 2025

- GPA: 3.73/4.0 | Relevant Coursework: Machine Learning, NLP, Data Science, Data Mining, Data Visualization, Data Engineering, Database Management, Digital Image Processing, Data Analytics

Malla Reddy Engineering College, Bachelor of Technology in Computer Science

July 2021

- GPA: 9.36/10.0 | Relevant Coursework: Algorithms, Data Structures, Operating Systems, Databases, Machine Learning, Computer Networks, Software Engineering

Certifications & Achievements

- AWS Certified Solutions Architect – Associate (Amazon Web Services), June 2024; Google Cloud Specialization: Architecting with Google Compute Engine, Google Certified - Generative AI for Educators (2025)
- “*Human-in-the-Loop Safety for MCP Database Servers*”, *IEEE TechRxiv (Preprint)*, 2025. DOI: 10.36227/techrxiv.175979372.21006002/v1.
- Certified in Java, Python, and Apache Airflow (HackerRank, Udemy) and in C and Data Structures (QEEE by IIT-Madras).
- Semi-Finalist in TechGig Code Gladiators 2020 (Top 0.1%, outperforming 250k participants).
- Open-source contributor during HacktoberFest (2019, 2022) and completed virtual experiences with JP Morgan and Goldman Sachs.
- Recognized with the Experian Spot Award for driving impactful outcomes and collaboration with product stakeholders.